Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The reaction $\mathrm{CH_3COCH_3 + 4H \xrightarrow[\text{conc. HCl}]{\text{Zn-Hg}} C_3H_8 + H_2O}$ is known as

(A) Rosenmund's reduction

(B) Clemmensen reduction

(C) Wurtz reaction

(D) Kolbe's reaction

Q.2

Which of the following is expected to have minimum boiling point?

(A) n-Pentane

(B) n-Butane

(C) 2-Methylbutane

(D) 2, 2-Dimethylpropane

Q.3

The most strained cycloalkane is

(A) cyclopropane

(B) cyclobutane

(C) cyclopentane

(D) cyclohexane

Q.4

Ethanol on treatment with concentrated HI and red phosphorus forms

(A) $\mathrm{C_2H_5I}$

(B) $\mathrm{C_2H_4}$

(C) $\mathrm{C_2H_6}$

(D) $\mathrm{C_3H_8}$

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Q.5

During debromination of meso-dibromobutane, the major compound formed is

(A) n-butane

(B) 1 -butene

(C) cis-2-butene

(D) trans-2-butene

Q.6

Ethylene when oxidized with hot acidified dichromate solution gives

(A) ethylene glycol

(B) acetic acid

(C) ethyl alcohol

(D) carbon dioxide and water

Q.7

The ozonolysis of 2,3 -dimethylpent-2-ene produces

(A) two molecules of aldehydes

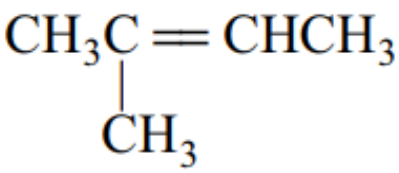
(B) two molecules of ketones

(C) one molecule of aldehyde and one molecule of ketone

(D) neither aldehyde nor ketone

Q.8

The treatment of



with $NaIO_4$ or boiling $KMnO_4$ produces

(A) CH_3COCH_3 and CH_3COOH

(B) CH_3COCH_3 and $HCHO$

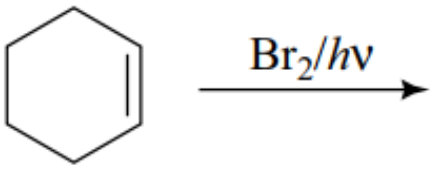
(C) CH_3CHO and CO_2

(D) CH_3COCH_3 only

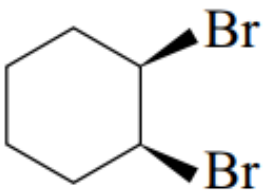
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Q.9

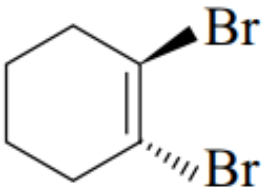
Bromination of cyclohexene under conditions given below yields:



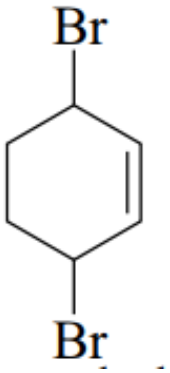
(A)



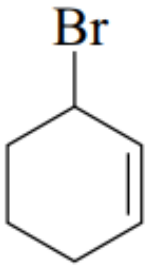
(B)



(C)



(D)



Q.10

Which of the following has the highest boiling point?

(A) 1-Butyne

(B) 2-Butyne

(C) 1-Pentyne

(D) 2-Hexyne

Q.11

The hydrolysis of Mg_2C_3 produces

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(A) acetylene

(B) propyne

(C) butyne

(D) ethylene

Q.12

The product(s) obtained via oxymercuration $(HgSO_4 + H_2SO_4)$ of 1-butyne would be

(A) $CH_3CH_2COCH_3$

(B) $CH_3CH_2CH_2CHO$

(C) $CH_3CH_2CHO + HCHO$

(D) $CH_3CH_2COOH + HCOOH$

Q.13

The addition of H_2 to $CH_3C \equiv CCH_3$ using Lindlar's catalyst produces

(A) cis-but-2-ene

(B) trans-but-2-ene

(C) a mixture of cis- and trans-but-2-ene

(D) 2-methylprop-1-ene

Q.14

Ozonolysis of propyne gives

(A) acetic acid only

(B) formic acid only

(C) acetic and formic acids

(D) acetaldehyde

Q.15

Which of the following compounds reacts with an aqueous solution of $Ag(NH_3)OH$?

(A) 2-Butyne

(B) 1 -Butyne

(C) Ethene

(D) Ethane

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Q.16

The cyclic polymerization of methyl acetylene produces

(A) benzene

(B) *o*-xylene

(C) 1, 3, 5-trimethylbenzene

(D) 1,3,5-trimethyl cyclohexane

Q.17

A direct iodination of benzene is not possible because

(A) iodine is an oxidizing agent

(B) resulting C_6H_5I is reduced to C_6H_6 by HI

(C) HI is unstable

(D) the ring gets deactivated

Q.18

The major product when $1\ mol$ of H_2 is added to $C_6H_5CH=CHCH=CHCH_3$ using Pt catalyst is

(A) $C_6H_5CH_2CH=CHCH_2CH_3$

(B) $C_6H_5CH_2CH_2CH=CHCH_3$

(C) $C_6H_5CH=CHCH_2CH_2CH_3$

(D) $C_6H_5CH_2CH_2CH_2CH=CH_2$

Q.19

In the Friedel-Crafts acylation, the electrophile is

(A) $C_6H_5^+$

(B) $AlCl_3^-$

(C) CH_3CO^+

(D) $C_6H_5CH_2^+$

Q.20

Among the following, the compound that can be readily sulphonated is

(A) benzene

(B) toluene

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(C) nitrobenzene

(D) benzaldehyde

Q.21

0.46 g of a compound with molecular mass of 92 g gave 336 mL of a gas at STP when treated with excess of CH_3MgI. The number of moles of gas evolved or the number of active H atoms in the compound is:

Q.22

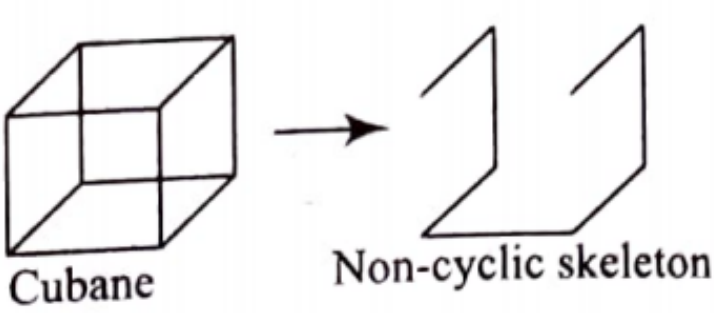
The addition of 4.12mg of an unknown alcohol, ROH, to CH_3MgI releases 1.56 mL of a gas at STP. The molar mass of alcohol is (round off to the nearest integer):

Q.23

A sample of diesel has the same characteristics as a 60 mL mixture of cetane and α-methyl napthalene mixed in $2:1$ ratio (v/v). What is the cetane number of the diesel sample? (round off to the nearest integer)

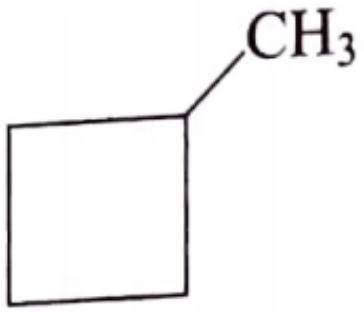
Q.24

How many bond cleavages are required to convert cubane into non-cyclic skeleton?



Q.25

How many distinct monobromination products, including stereoisomers, can result when the alkane below is heated in the presence of Cl_2?



Answers & Solutions

Q.1 Answer:

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Clemmensen reduction

Solution:

This reaction is known as Clemmensen reduction.
In this reaction, $C = O$ group is reduced to CH_2 group.
This reaction is useful for carbonyl compounds which are sensitive to base.

Q.2 Answer:

n-Butane

Solution:

The boiling point is directly proportional to molecular mass. So n-butane among given will have a minimum boiling point.

Q.3 Answer:

cyclopropane

Solution:

In cyclopropane, the $C - C - C$ bond angles are 60° whereas tetrahedral 109.5° bond angles are expected.
The intense angle strain leads to nonlinear orbital overlap of its sp^3 orbitals.
Because of the bond's instability, cyclopropane is more reactive than other alkanes. Since any three points make a plane and cyclopropane has only three carbons, cyclopropane is planar.
The $H - C - H$ bond angle is 115° whereas 106° is expected as in the CH_2 groups of propane.

Q.4 Answer:

C_2H_6

Solution:

$$C_2H_5OH \xrightarrow{HI/P} C_2H_6 + H_2O$$

Q.5 Answer:

trans-2-butene

Solution:

In the debromination reaction of meso-dibromobutane, the two bromine atoms leave the molecule and to stabilise the compound forms a double bond. The two products namely cis-but$-2-$ene and $trans - but - 2 - ene$ formed in the reaction. As trans$-2-$butene is more stable than cis-but $-2-$ene, it is the major product.

Q.6 Answer:

carbon dioxide and water

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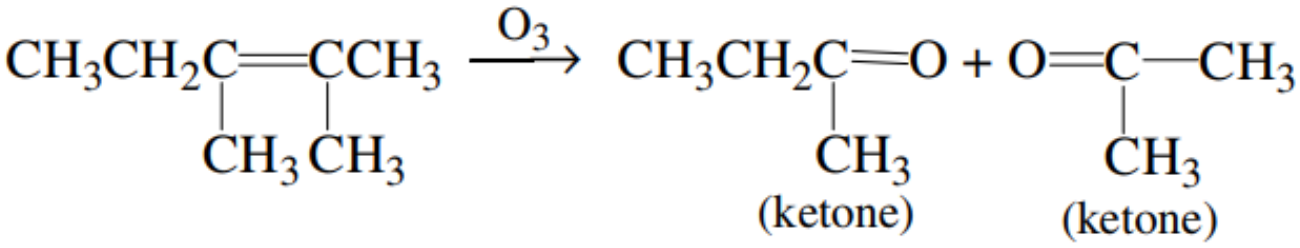
Solution:

$$CH_2 = CH_2 \xrightarrow{Cr_2O_7^{2-}/H^+} 2CO_2 + 2H_2O$$

Q.7 Answer:

two molecules of ketones

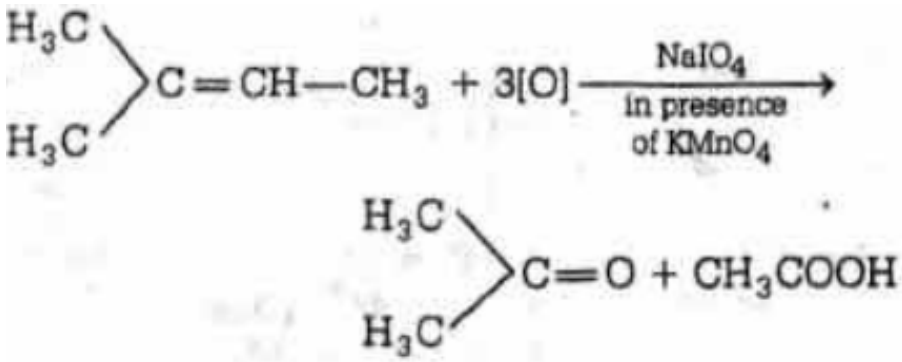
Solution:



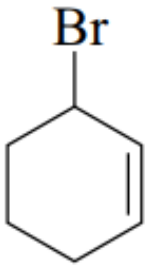
Q.8 Answer:

CH_3COCH_3 and CH_3COOH

Solution:

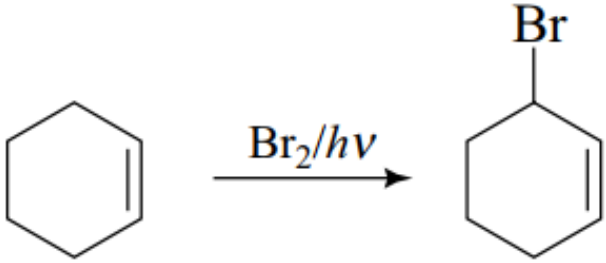


Q.9 Answer:



Solution:

In the presence of ultraviolet radiation, the allylic $C-H$ bond undergoes bromination.



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Q.10 Answer:

2-Hexyne

Solution:

The boiling point increases with increase in carbon number.

Q.11 Answer:

propyne

Solution:

$$Mg_2C_3 + 4H_2O \rightarrow CH_3C \equiv CH + 2Mg(OH)_2$$

Q.12 Answer:

$CH_3CH_2COCH_3$

Solution:

$$CH_3CH_2C \equiv CH \xrightarrow[2H_2O]{HgSO_4/H_2SO_4} CH_3CH_2C(OH)_2CH_3 \xrightarrow{-H_2O} CH_3CH_2\overset{\overset{O}{\|}}{C}CH_3$$

$1-$butyne on reaction with mercurous sulphide and sulphuric acid forms isobutyl alcohol as a product which undergoes tautomerism to form ethyl methyl ketone as a product.

Q.13 Answer:

cis-but-2-ene

Solution:

The use of Lindlar's catalyst causes cis-addition.

Q.14 Answer:

acetic and formic acids

Solution:

$$CH_3C \equiv CH \xrightarrow{O_3} CH_3COOH + HCOOH$$

Q.15 Answer:

1 -Butyne

Solution:

A compound containing acidic hydrogen in $-C \equiv CH$ group reacts with $Ag(NH_3)OH$.

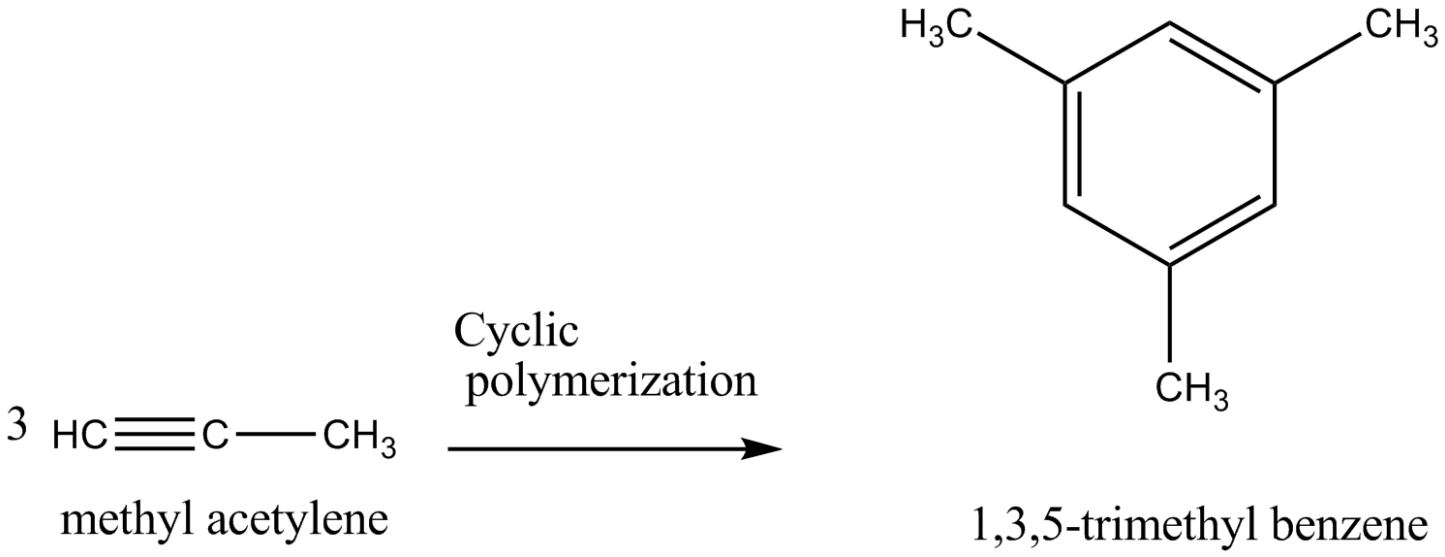
Q.16 Answer:

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1, 3, 5-trimethylbenzene

Solution:

The cyclic polymerization of methyl acetylene produces $1, 3, 5$-trimethyl benzene. Three molecules of methyl acetylene polymerize to produce 1 molecule of $1, 3, 5$-trimethyl benzene.



Q.17 Answer:

resulting C_6H_5I is reduced to C_6H_6 by HI

Solution:

C_6H_5I is reduced by HI to C_6H_6

Q.18 Answer:

$C_6H_5CH = CHCH_2CH_2CH_3$

Solution:

Double bond attached to methyl group is reduced.

Q.19 Answer:

CH_3CO^+

Solution:

The electrophile is CH_3CO^+.

Q.20 Answer:

toluene

Solution:

$-CH_3$ group in toluene activates benzene ring, its sulphonation is easier than rest of the three molecules.

Q.21 Answer:

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Solution:

At STP, volume of $1\ \text{mol}$ **of a gas** $= 22.4$ **litres (** **when** $P = 1$ **atm,** $T = 273\ \text{K}$**). If** $P = 1$ **bar and** $T = 273\ \text{K}$**, then volume** $= 22.7$ **litres. Similarly, at standard condition or SATP when** $P = 1\ \text{atm}$ **and** $T = 298\ \text{K}$**, then volume** $= 24.4$ **litres. When** $P = 1$ **bar and** $T = 298\ \text{K}$**, then volume** $= 24.7$ **litres.**

$0.46\ \text{g}$ **of a compound** $\Rightarrow 336\ \text{mL}$ **of a gas at STP.**

$$\therefore 92\ \text{g of a compound} \Rightarrow \frac{336 \times 92}{0.46} = 67200\ \text{mL}$$

$$= \frac{67200}{22400} = 3\ \text{mol}$$

Q.22 Answer:

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Solution:

At STP, volume of $1\ \text{mol}$ **of a gas** $= 22.4$ **litres (** **when** $P = 1$ **atm,** $T = 273\ \text{K}$**). If** $P = 1$ **bar and** $T = 273\ \text{K}$**, then volume** $= 22.7$ **litres. Similarly, at standard condition or SATP when** $P = 1\ \text{atm}$ **and** $T = 298\ \text{K}$**, then volume** $= 24.4$ **litres. When** $P = 1$ **bar and** $T = 298\ \text{K}$**, then volume** $= 24.7$ **litres.**

$1.56\ \text{mL}$ **of a gas is obtained from** $4.12 \times 10^{-3}\ \text{g}$ **of** ROH.

Therefore, $22400\ \text{mL}$ **of a gas is obtained from**

$$\frac{4.12 \times 10^{-3} \times 22400}{1.56} = 59.15\ \text{g mol}^{-1}$$

Q.23 Answer:

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Solution:

Sixty millilitres of a mixture contains $40\ \text{mL}$ **cetane and** $20\ \text{m}L$ **of** α**-methylnaphthalene.**

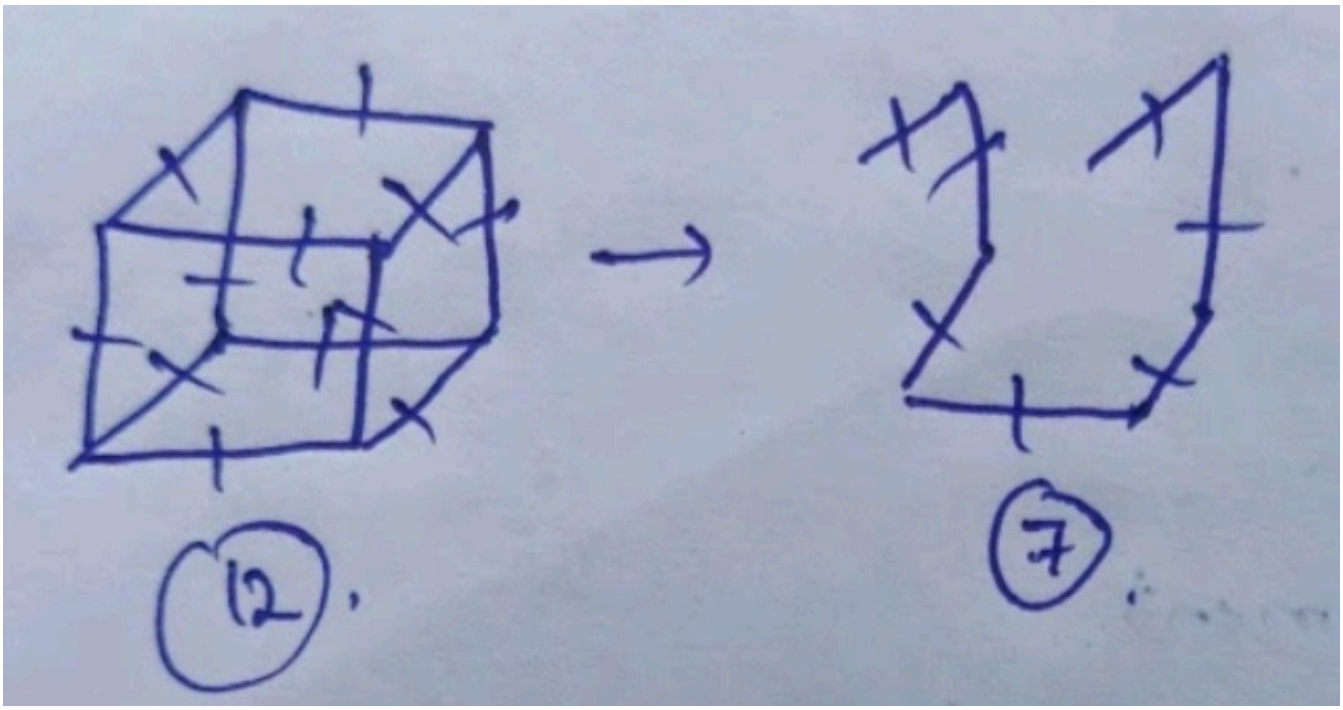
Therefore, cetane number $= \frac{40 \times 100}{60} = 66.6\%$

Q.24 Answer:

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Solution:

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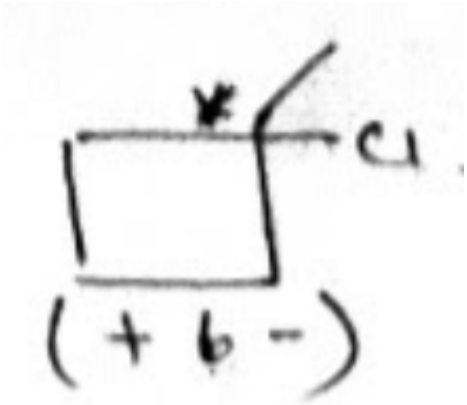
So, 5 cleavages are reqired.

Q.25 Answer:

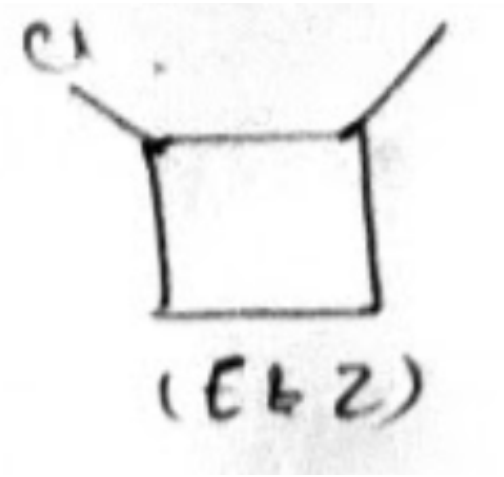
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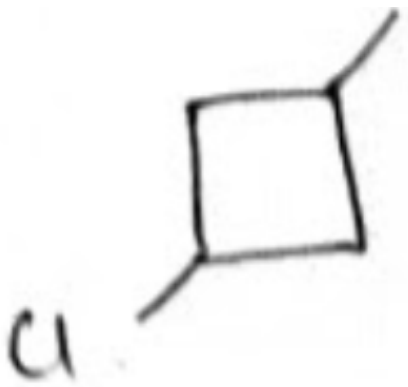
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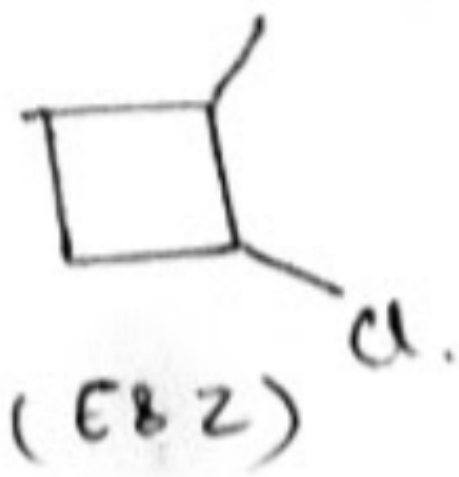




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∴ **Total 8 monobromination products.**

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